

HCSM-40

The Higgs Mechanism

The Higgs Doublet as the Complexified B_1 Isotypic, the Vacuum Weak Direction from the Zero State, and the Two Loop Anomaly Closure

Hodge Complex Standard Model

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September 2026

Abstract

We derive the Higgs doublet of the Hodge Complex Standard Model (HCSM) as the complexification of the B_1 isotypic of $\Omega^0|_{\lambda=4}$, and we derive the physical weak direction from the vacuum zero state s . The Higgs doublet lives in the 2-dimensional real B_1 multiplicity space; its complexification is the Standard Model Higgs doublet with the correct $SU(2)_L$ action. The zero state fixes the weak Cartan with eigenvalues ± 1 . The DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ preserves $\mathfrak{su}(2)_L$ exactly for every interaction strength U by a doubled momentum cancellation mechanism. We close the two loop anomaly on the Hodge complex by direct computation: the chirality grading $\gamma_5 = (-1)^F$ anticommutes with the Hodge-Dirac operator, forcing $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f , and the one loop Standard Model content satisfies $\text{Tr}(Y) = \text{Tr}(Y^3) = SU(2)^2 U(1) = SU(3)^2 U(1) = 0$ in exact rational arithmetic.

The tree level quartic coupling and the top-Yukawa radiative correction are reported as numerical candidates, not derivations. The formula $\lambda_{\text{tree}} = \ln(kL)/(3\pi^2)$ matches observation to 0.1% given the calibration anchor v_{EW} , and the formula $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ matches the observed top Yukawa to 0.002%. Both candidates use only HCSM-native constants, but neither is derived from the substrate dynamics. The origin of the coefficient 4/3 in the top-Yukawa formula is not identified. The electroweak scale v_{EW} is a calibration anchor and is not derived from the framework.

Derivation status summary.

- **DERIVED:** the Higgs doublet as the complexified B_1 isotypic; the weak direction from the zero state s ; the exact $\mathfrak{su}(2)_L$ preservation by the DNLS Hartree Hamiltonian; two-loop anomaly cancellation on the Hodge complex.
- **DISCLOSED INPUTS:** the Standard Model per-generation content; the physical identification $A_1 \leftrightarrow \text{color}$ and $B_1 \leftrightarrow \text{weak isospin}$; the sector assignment $u_R^c \leftrightarrow \Omega^0$, $Q_L \leftrightarrow \Omega^1$, $d_R^c \leftrightarrow \Omega^2$; the electroweak anchor $v_{\text{EW}} = 246.21965 \text{ GeV}$.
- **OPEN:** a first-principles derivation of the tree-level quartic coupling λ_{tree} from the Hodge complex; a first-principles derivation of the top-Yukawa correction $\Delta\lambda_{\text{top}}$ from substrate dynamics; a derivation of the electroweak scale v_{EW} (candidate no-go).

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1 Introduction

1.1 The Higgs problem

The Standard Model Higgs sector is parameterized by two quantities: the electroweak vacuum expectation value $v_{\text{EW}} = 246.21965$ GeV and the quartic coupling $\lambda \approx 0.129$, from which the Higgs mass $m_H = \sqrt{2\lambda} v_{\text{EW}} = 125.25$ GeV follows. Neither quantity is derived from first principles in the Standard Model; both are empirical anchors.

The Hodge Complex Standard Model (HCSM) derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, the per-generation matter content, and the Standard Model hypercharges from the Hodge complex of differential forms on the 8×8 torus (HCSM-06, HCSM-07, HCSM-08). The 56-dimensional $\lambda = 4$ sector decomposes under the D_4 point group as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$. The color algebra $\mathfrak{su}(3)_c$ acts on the A_1 isotypic; the weak isospin algebra $\mathfrak{su}(2)_L$ acts on the B_1 isotypic (HCSM-06).

This paper addresses the Higgs sector. We derive the Higgs doublet as the complexification of the B_1 isotypic of $\Omega^0|_{\lambda=4}$, and we derive the physical weak direction from the vacuum zero state s . We then compute the two-loop anomaly on the Hodge complex and close it with a structural derivation. The tree-level quartic and the top-Yukawa correction are reported as numerical candidates with HCSM-native inputs, but their derivation from substrate dynamics remains open.

1.2 Summary of results

1. (Theorem 3.2). The Higgs doublet is the complexification of the B_1 isotypic of $\Omega^0|_{\lambda=4}$.
2. (Theorem 4.2). The physical weak Cartan is fixed by the vacuum zero state s and has eigenvalues ± 1 on the B_1 subspace.
3. (Theorem 5.2). The DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ preserves $\mathfrak{su}(2)_L$ exactly, for every $U \in \mathbb{R}$, by a doubled-momentum cancellation mechanism.
4. (Theorem 6.5). Two-loop anomaly cancellation on the Hodge complex is exact: the chirality grading satisfies $\{\gamma_5, D\} = 0$, forcing $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator, and the one-loop Standard Model content has $\text{Tr}(Y) = \text{Tr}(Y^3) = \text{SU}(2)^2\text{U}(1) = \text{SU}(3)^2\text{U}(1) = 0$.
5. (Proposition 7.1). The quartic coupling $\lambda_{\text{tree}} = \ln(kL)/(3\pi^2)$ matches the observed Higgs mass to 0.09% given the calibration anchor v_{EW} . The formula uses the HCSM-native warp factor $kL = 38.442527$ but is not derived from the Hodge complex.
6. (Proposition 8.1). The top Yukawa satisfies the candidate formula $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ to 0.002% accuracy, where $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$. All inputs are HCSM-native, but the origin of the coefficient $4/3$ is not identified.

1.3 Organization

Section 2 recalls the Hodge complex, the 14-mode multiplet, the B_1 isotypic, and the zero state. Section 3 derives the Higgs doublet. Section 4 derives the physical weak direction. Section 5 proves the DNLS preservation of $\mathfrak{su}(2)_L$. Section 6 closes the two-loop anomaly. Sections 7 and 8 report the quartic and top-Yukawa candidates with full honest accounting. Section 9 catalogues what is derived, what is disclosed, and what is open. Section 10 states falsifiable predictions. Appendix A provides reproducible verification.

2 Setup

2.1 The Hodge complex

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The Hodge complex is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (1)$$

with total dimension 256. The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree (HCSM-02).

2.2 The 14-mode multiplet and B_1 isotypic

The self-paired eigenvalue $\lambda = 4$ of the discrete 5-point Laplacian has multiplicity 14 on Ω^0 , realized on the mode set

$$\mathcal{M} = \mathcal{O}_0 \cup \mathcal{O}_1 \cup \mathcal{O}_2, \quad (2)$$

with

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\} \quad (3)$$

The D_4 decomposition is

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (4)$$

The B_1 irrep is 1-dimensional over $\mathbb{R}$. Its multiplicity in $\Omega^0|_{\lambda=4}$ is 2. The B_1 isotypic is therefore a real 2-dimensional subspace of the 14-mode multiplet.

2.3 The zero state

The 14×19 measurement matrix M of vacuum-native observables has rank 13 and nullity 1. Its unique nullspace vector s , normalized to $\max_i |s_i| = 1$, is the zero state (HCSM-22). The components relevant to the electroweak sector are

$$s_H = +0.329142, \quad s_t = 0, \quad s_\nu = -0.105198, \quad s_e = +0.105198. \quad (5)$$

2.4 Disclosed inputs

This paper uses three disclosed inputs from HCSM-06, stated explicitly in the framework's rules:

1. **Input 1 (SM content).** The Standard Model per-generation content: $Q_L(3, 2, 1/6) \times 6$, $u_R^c(3, 1, -2/3) \times 3$, $d_R^c(3, 1, 1/3) \times 3$, $L_L(1, 2, -1/2) \times 2$, $e_R^c(1, 1, 1) \times 1$, $\nu_R^c(1, 1, 0) \times 1$.
2. **Input 2 (Physical identification).** The identification of the A_1 isotypic with physical color and the B_1 isotypic with physical weak isospin.
3. **Input 3 (Sector assignment).** The assignment of u_R^c , Q_L , d_R^c to the Hodge sectors Ω^0 , Ω^1 , Ω^2 respectively.

Additionally, the electroweak vacuum expectation value $v_{EW} = 246.21965$ GeV is used as a calibration anchor in Sections 7 and 8. This is disclosed per HCSM-00 rules and its role is analysed in Section 9.

3 The Higgs Doublet as the Complexified B_1 Isotypic

Definition 3.1 (B_1 isotypic). *Let $V_{B_1} \subset \Omega^0|_{\lambda=4}$ be the isotypic subspace of the B_1 irrep. Then $\dim_{\mathbb{R}} V_{B_1} = 2$.*

Theorem 3.2 (Higgs doublet). *The complexification $V_{B_1} \otimes_{\mathbb{R}} \mathbb{C}$ is a 2-dimensional complex vector space carrying the fundamental representation of $\mathfrak{su}(2)_L$. It is the Standard Model Higgs doublet.*

Proof. The B_1 multiplicity in $\Omega^0|_{\lambda=4}$ is 2 from (4). The $\mathfrak{su}(2)_L$ generators act on the B_1 isotypic via the Pauli matrices (HCSM-06, Theorem 5.2), giving the fundamental representation on $\mathbb{C}^2$. The complex doublet $H = (H^+, H^0)$ is the basis of this representation. The hypercharge assignment $Y(H) = 1/2$ follows from the sector-coarse hypercharge construction (HCSM-07). $\square$

Remark 3.3. *The Higgs doublet is not introduced as a new field. It is the B_1 isotypic of the Hodge complex, which already appears in the gauge-algebra derivation. The complexification is the passage from the real 2-dimensional D_4 -invariant subspace to the complex representation space of $\mathfrak{su}(2)_L$.*

4 The Physical Weak Direction from the Zero State

Definition 4.1 (Aligned Cartan). *Let $\{w_1, w_2\}$ be an orthonormal basis of V_{B_1} . The vacuum zero state s has projection $s_{B_1} = \sum_a s_a w_a$. Define the aligned Cartan*

$$T_3^{\text{aligned}} = \hat{n} \cdot \vec{T}, \quad \hat{n} = \frac{s_{B_1}}{\|s_{B_1}\|}, \quad (6)$$

where $\vec{T} = (T_L^1, T_L^2, T_L^3)$ are the $\mathfrak{su}(2)_L$ generators.

Theorem 4.2 (Weak direction). *The aligned Cartan T_3^{aligned} has eigenvalues ± 1 on V_{B_1} , and defines the physical weak up/down direction.*

Proof. The $\mathfrak{su}(2)_L$ generators are $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ with σ^i the Pauli matrices and U_{B_1} the basis change to V_{B_1} (HCSM-06, Theorem 5.2). The Cartan $T_3^{\text{aligned}} = \hat{n} \cdot \vec{T}$ is a unit-norm element of the $\mathfrak{su}(2)$ algebra. Any unit-norm element of $\mathfrak{su}(2)$ has eigenvalues ± 1 in the fundamental representation. Direct numerical diagonalization gives ± 1 to machine precision. $\square$

Remark 4.3. *The zero state s is the unique vacuum configuration determined by the 14×19 measurement matrix. The weak direction is therefore not a free parameter: it is fixed by the vacuum.*

5 DNLS Preservation of $\mathfrak{su}(2)_L$

Definition 5.1 (DNLS Hartree Hamiltonian). *The DNLS Hartree Hamiltonian on the 14-mode multiplet is*

$$H_{\text{DNLS}} = 4\mathbb{I} + UK, \quad K[a, b] = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (7)$$

where U is the on-site interaction strength.

Theorem 5.2 (Exact $\mathfrak{su}(2)_L$ preservation). *For every $U \in \mathbb{R}$, the commutator $[H_{\text{DNLS}}, T_L^i] = 0$ for all three generators T_L^i of $\mathfrak{su}(2)_L$.*

Proof sketch. By the doubled-momentum pairing lemma (HCSM-03, Lemma 7), the B_1 basis vectors v_0 and w_1 pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ with opposite signs. This makes the compressed Hartree matrix $K_{B_1} = U_{B_1}^T K U_{B_1}$ vanish identically on V_{B_1} . Since the $\mathfrak{su}(2)_L$ generators are supported entirely on V_{B_1} , the commutator $[K, T_L^i] = 0$. Full proof in HCSM-03. $\square$

Corollary 5.3. *The DNLS dynamics preserves the weak isospin structure of the substrate for any interaction strength.*

6 Two-Loop Anomaly Cancellation on the Hodge Complex

6.1 The chirality grading

Definition 6.1 (Chirality grading). *The chirality grading on the Hodge complex is*

$$\gamma_5 = (-1)^F = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}. \quad (8)$$

It assigns eigenvalue +1 to even forms $(\Omega^0 \oplus \Omega^2)$ and -1 to odd forms (Ω^1) .

Theorem 6.2 (Anticommutation). *$\{\gamma_5, D\} = \gamma_5 D + D\gamma_5 = 0$ exactly.*

Proof. The Hodge-Dirac operator is odd with respect to the $\mathbb{Z}_2$ grading: it maps even forms to odd forms and odd forms to even forms (HCSM-02). Therefore $\gamma_5 D = -D\gamma_5$, which is equivalent to the anticommutation relation. Direct numerical verification gives $\|\gamma_5 D + D\gamma_5\|_F = 0$ exactly in IEEE-754 double precision. $\square$

Corollary 6.3 (Regulator trace vanishes). *For any even function f with $f(\lambda) = f(-\lambda)$,*

$$\text{Tr}(\gamma_5 f(D)) = 0. \quad (9)$$

Proof. The spectrum of D is $\pm$ -symmetric within each sector (HCSM-02, Theorem 6.1). The eigenvalues pair as $\lambda \leftrightarrow -\lambda$ with opposite chirality. Therefore the trace of $\gamma_5 f(D)$ vanishes identically for every even f . Numerical verification across nine regulators spanning compact support, exponential decay, power decay, and oscillatory forms gives residuals at the level of machine epsilon (10^{-15}). $\square$

Corollary 6.4 (Index of D). *$\text{Index}(D) = \text{Tr}(\gamma_5 P_{\ker D}) = 0$.*

Proof. The Betti numbers of the torus are $(b_0, b_1, b_2) = (1, 2, 1)$ (HCSM-02). The index of D is $b_0 - b_1 + b_2 = 0$ (the Euler characteristic of T^2). Direct numerical verification gives $\text{Tr}(\gamma_5 P_{\ker D}) = 0$ at machine precision. $\square$

6.2 The two-loop anomaly

Theorem 6.5 (Two-loop anomaly cancellation). *The two-loop anomaly on the Hodge complex vanishes exactly.*

Proof. The two-loop anomaly coefficient on a lattice factorizes as

$$A^{(2)} = \text{Tr}_{\text{Hodge}}(\gamma_5 R(D)) \cdot \text{Tr}_{\text{content}}(Y^3), \quad (10)$$

where $R(D)$ is the regulator. Each factor vanishes independently:

Factor 1 (geometry). By Corollary 6.3, $\text{Tr}(\gamma_5 R(D)) = 0$ for every even regulator R . This is exact and does not depend on the fermion content.

Factor 2 (content). The Standard Model per-generation content has

$$\text{Tr}(Y) = 6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 \cdot 1 + 1 \cdot 0 = 0, \quad (11)$$

$$\text{Tr}(Y^3) = 0, \quad (12)$$

$$\text{SU}(2)^2 \text{U}(1) = 0, \quad (13)$$

$$\text{SU}(3)^2 \text{U}(1) = 0, \quad (14)$$

verified in exact rational arithmetic.

Both factors vanish. Their product is zero. The two-loop anomaly cancels by two independent mechanisms. $\square$

Remark 6.6 (Comparison with Adler-Bardeen). *The Adler-Bardeen theorem states that the ABJ anomaly is one-loop exact: if the one-loop anomaly vanishes, all higher-loop anomalies vanish too. The HCSM result is stronger: the Hodge-Dirac anticommutation $\{\gamma_5, D\} = 0$ is a geometric identity that forces the regulator trace to vanish at every loop order, independent of Adler-Bardeen. The two-loop cancellation is therefore a structural theorem of the Hodge complex, not an inherited property of the Standard Model content.*

7 The Tree-Level Quartic Coupling: Candidate Formula

7.1 The candidate

Proposition 7.1 (Quartic candidate). *The tree-level quartic coupling satisfies the candidate formula*

$$\lambda_{\text{tree}} = \frac{\ln(kL)}{3\pi^2}, \quad (15)$$

with $kL = 38.442527$ (HCSM-51), giving $\lambda_{\text{tree}} = 0.123246$.

7.2 Numerical consistency

Given the calibration anchor $v_{\text{EW}} = 246.21965$ GeV and the Standard Model 1-loop top-Yukawa correction $\Delta\lambda_{\text{top}} = (3y_t^4/16\pi^2) \ln(m_t/m_H)$, the total effective quartic is

$$\lambda_{\text{eff}} = \lambda_{\text{tree}} + \Delta\lambda_{\text{top}} = 0.123246 + 0.005906 = 0.129152. \quad (16)$$

The implied Higgs mass is

$$m_H = \sqrt{2\lambda_{\text{eff}}} \cdot v_{\text{EW}} = 125.138 \text{ GeV}, \quad (17)$$

matching the observed $m_H = 125.25 \pm 0.17$ GeV to 0.090%, with a statistical pull of 0.66σ .

7.3 Honest status

The formula $\lambda_{\text{tree}} = \ln(kL)/(3\pi^2)$ is a 5D holographic ansatz. It uses the HCSM-native warp factor kL but is not derived from the Hodge complex. The numerical match to the observed Higgs mass is 0.09%, but this is achieved by combining the candidate formula with the calibration anchor v_{EW} and the Standard Model 1-loop correction.

The specific coefficient $3\pi^2$ in the denominator is the 4D loop factor with $N_c = 3$. It is natural but not derived. The $1/(3\pi^2)$ factor is what converts the dimensionless warp depth $\ln(kL)$ into a dimensionless quartic coupling.

Status: candidate formula, numerically consistent, not derived.

8 The Top-Yukawa Radiative Correction: Candidate Formula

8.1 The candidate

Proposition 8.1 (Top-Yukawa candidate). *The top Yukawa satisfies the candidate formula*

$$y_t^2 = 1 - \frac{4}{3}\delta_{\text{edge}}, \quad (18)$$

where $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$ is the class-A DME residual (HCSM-20, Theorem 6.5).

8.2 Numerical consistency

The inputs are HCSM-native:

$$\gamma = 0.70283946799007245929 \dots \quad (\text{HCSM-54}), \quad (19)$$

$$\delta_{\text{edge}} = \sqrt{2}/\gamma - 2 = 0.012143066491921 \dots \quad (20)$$

The formula gives

$$y_t^2 = 1 - \frac{4}{3} \cdot 0.012143066491921 = 0.9838092447, \quad y_t = 0.991872. \quad (21)$$

The observed value is $y_t^{\text{obs}} = \sqrt{2} \cdot 172.69/246.21965 = 0.991881$. Residual: 0.002% on y_t^2 , 0.001% on y_t .

For comparison, the direct SM 1-loop correction is $\Delta\lambda_{\text{top}} = 0.005906$ with y_t^{obs} . Using y_t from the candidate formula: $\Delta\lambda_{\text{top}}^{\text{HCSM}} = 0.0059061$, matching to 0.002%.

8.3 The structural input: the top's unique position

The top mode $t = (6, 6)$ sits at the intersection of four structural features of the Hodge complex:

1. It is in orbit $\mathcal{O}_2$ (support 32).
2. It is in loop class A ($\omega_A = 2\sqrt{2}$).
3. Its zero-state amplitude $s_t = 0$ exactly, forced by the diagonal reflection $(n_1, n_2) \rightarrow (n_2, n_1)$.
4. It is a fixed point of the diagonal reflection, along with $(2, 2)$.

No other charged fermion has all four features. The top is therefore structurally singled out.

8.4 The coefficient $4/3$

The coefficient $4/3$ admits three natural interpretations that coincide:

$$\frac{4}{3} = \frac{d}{d-1} = \frac{|\mathcal{O}_2|}{d-1} = \frac{2d-4}{d-1}. \quad (22)$$

All three are HCSM-native. The first is the dimension ratio, the second is the orbit-size-to-generation-count ratio, and the third is an equivalent algebraic form. However, the best-fit coefficient from the observed top Yukawa is $k = 1.331830$, which differs from $4/3$ by 0.11%. The candidate is not exact.

8.5 Honest status

The formula $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ uses only HCSM-native quantities. It matches observation to 0.002%, which is $180\times$ smaller than the experimental uncertainty on y_t propagated through $m_t = 172.69 \pm 0.30$ GeV. But the specific origin of the coefficient $4/3$ from substrate dynamics is not identified. The formula does not generalize to the other charged fermions in any obvious way: for $u, d, s, c, b, \tau, \mu, e$, the effective coefficient $k = (1 - y_f^2)/\delta_{\text{edge}}$ takes the trivial value $1/\delta_{\text{edge}} \approx 82.35$ because $y_f \ll 1$. The top is uniquely singled out.

Status: candidate formula, numerically consistent, not derived.

9 Derivation Status

9.1 Derived from the substrate

1. The Higgs doublet is the complexification of the B_1 isotypic of $\Omega^0|_{\lambda=4}$ (Theorem 3.2).
2. The physical weak direction is fixed by the vacuum zero state s and has eigenvalues ± 1 (Theorem 4.2).
3. The DNLS Hartree Hamiltonian preserves $\mathfrak{su}(2)_L$ exactly, for every $U \in \mathbb{R}$ (Theorem 5.2).
4. The chirality grading anticommutes with the Hodge-Dirac operator (Theorem 6.2).
5. The regulator trace vanishes for every even function (Corollary 6.3).
6. The index of D is zero (Corollary 6.4).
7. Two-loop anomaly cancellation on the Hodge complex is exact (Theorem 6.5).

9.2 Disclosed inputs

1. Standard Model per-generation content (Input 1).
2. Physical identification of A_1 with color and B_1 with weak isospin (Input 2).
3. Sector assignment $u_R^c \leftrightarrow \Omega^0$, $Q_L \leftrightarrow \Omega^1$, $d_R^c \leftrightarrow \Omega^2$ (Input 3).
4. Electroweak anchor $v_{\text{EW}} = 246.21965$ GeV, used in Sections 7 and 8.

9.3 Numerically consistent candidates (not derived)

1. The tree-level quartic $\lambda_{\text{tree}} = \ln(kL)/(3\pi^2)$, matching the observed Higgs mass to 0.09% given v_{EW} (Proposition 7.1).
2. The top Yukawa $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$, matching observation to 0.002% (Proposition 8.1).

9.4 Open

1. **Derive λ_{tree} from the Hodge complex.** The formula $\ln(kL)/(3\pi^2)$ is a 5D holographic ansatz. A Hodge-native derivation would express the quartic coupling as a trace over the B_1 isotypic or as a one-loop correction on the torus. The specific origin of the $3\pi^2$ denominator and the $\ln(kL)$ numerator is not identified.
2. **Derive the coefficient $4/3$ in the top-Yukawa formula.** The best-fit coefficient from observation is $k = 1.331830$, which differs from $4/3$ by 0.11%. The exact value (if it exists) is not identified. A substrate-level derivation of the coefficient from the top's unique structural position is open.
3. **Derive v_{EW} from the framework.** The electroweak scale is a calibration anchor. A derivation from kL , γ , or other HCSM-native constants would require a dimensional-anchor mechanism that is not currently present. This is a candidate no-go: the framework may not have the structure to produce a dimensionful scale from the dimensionless constants alone.
4. **Derive the Higgs potential shape.** The Mexican-hat form $V(H) = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$ is assumed. Whether the Hodge complex forces this form (or a different one) is open.
5. **Derive $\Delta\lambda_{\text{top}}$ from substrate dynamics.** The 1-loop top-Yukawa correction is currently computed via the Standard Model formula. A HCSM-native computation from the DNLS mean-field or from a discrete loop integral on the torus is open.

10 Falsifiable Predictions

1. **Higgs doublet structure.** The Higgs doublet is the B_1 isotypic of $\Omega^0|_{\lambda=4}$. Any measurement that requires more than the 2-dimensional B_1 structure would falsify the derivation.
2. **Weak direction from zero state.** The weak Cartan is fixed by the vacuum. The two eigenvectors of T_3^{aligned} are the physical up/down weak states with eigenvalues ± 1 . A measurement of a different weak direction would falsify the theorem.
3. **DNLS preservation.** The DNLS Hartree Hamiltonian preserves $\mathfrak{su}(2)_L$ for every interaction strength U . A detection of $\mathfrak{su}(2)_L$ violation in the substrate interaction would falsify Theorem 5.2.
4. **Two-loop anomaly cancellation.** The Hodge complex gives exact two-loop cancellation by a structural mechanism independent of Adler-Bardeen. Any detection of a lattice anomaly at two loops would falsify Theorem 6.5.
5. **Quartic candidate.** $\lambda_{\text{tree}} = \ln(kL)/(3\pi^2)$ predicts $m_H = 125.138$ GeV given v_{EW} . A future refinement of v_{EW} or a different value of kL would test the formula.

6. **Top-Yukawa candidate.** $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ predicts $y_t = 0.991872$ given γ from HCSM-54. An improved measurement of m_t (or of γ via other observables) would test the formula at the 0.001% level.

11 Conclusion

We have derived the Higgs doublet as the complexified B_1 isotypic of $\Omega^0|_{\lambda=4}$, derived the physical weak direction from the vacuum zero state, and closed the two-loop anomaly on the Hodge complex by a structural mechanism that does not depend on Adler-Bardeen. The DNLS Hartree Hamiltonian preserves $\mathfrak{su}(2)_L$ exactly for every interaction strength.

The tree-level quartic coupling and the top-Yukawa radiative correction are reported as numerical candidates with HCSM-native inputs. Both match observation at the sub-percent level. Neither is derived from substrate dynamics. The origin of the coefficient $4/3$ in the top-Yukawa formula and the specific form $\ln(kL)/(3\pi^2)$ in the quartic formula remain open.

The electroweak scale $v_{\text{EW}} = 246.21965$ GeV is a calibration anchor. The framework does not currently derive it, and there is no obvious HCSM-native construction that would produce a dimensionful scale from the dimensionless constants $\{d, N, L, \gamma, kL\}$ alone. A candidate no-go is stated.

The paper’s contribution to the HCSM stack is threefold: the structural derivation of the Higgs doublet from the B_1 isotypic; the exact $\mathfrak{su}(2)_L$ preservation theorem for the DNLS Hartree Hamiltonian; and the two-loop anomaly closure on the Hodge complex. The remaining items are named explicitly and left for future work.

The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

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A Reproducible Verification

The companion script `HCSM-40_verification.py` verifies the following claims at machine precision:

1. The D_4 decomposition $\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$.
2. The B_1 isotypic has real dimension 2 and is the complexification basis for the Higgs doublet.
3. The zero state alignment: $s_t = 0$ exactly; the aligned Cartan has eigenvalues ± 1 .
4. The DNLS Hartree matrix compressed to V_{B_1} vanishes: $K_{B_1} = 0$ to machine precision.
5. The chirality grading anticommutation: $\|\{\gamma_5, D\}\|_F = 0$ exactly.
6. The regulator trace: $\text{Tr}(\gamma_5 f(D)) = 0$ for nine regulators (compact support, exponential, power, oscillatory).
7. Index of D : $\text{Tr}(\gamma_5 P_{\ker D}) = 0$.
8. Sector-resolved traces: $\text{Tr}_{\Omega^0}(e^{-D^2}) = \text{Tr}_{\Omega^2}(e^{-D^2}) = 6.0916487145$; $\text{Tr}_{\Omega^1}(e^{-D^2}) = 12.1832974290$; the γ_5 -weighted total vanishes.
9. One-loop anomaly coefficients: exact rational arithmetic gives $\text{Tr}(Y) = \text{Tr}(Y^3) = \text{SU}(2)^2\text{U}(1) = \text{SU}(3)^2\text{U}(1) = 0$.
10. Two-loop anomaly coefficient: $A^{(2)} = 0$.
11. Quartic candidate: $\lambda_{\text{tree}} = 0.123246$; $m_H = 125.138$ GeV; pull 0.66σ .
12. Top-Yukawa candidate: $y_t^2 = 1 - (4/3)\delta_{\text{edge}} = 0.9838092447$; residual from observation 1.83×10^{-5} .

Running `run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-40: The Higgs Mechanism
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; Higgs mechanism;
Higgs doublet; B1 isotypic; weak direction; zero state;
DNLS Hartree; $su(2)_L$ preservation; two-loop anomaly;
chirality grading; quartic coupling; top Yukawa
DC.description: We derive the Higgs doublet as the complexified
B1 isotypic of Ω^0 at $\lambda=4$, derive the physical weak
direction from the vacuum zero state, prove exact $su(2)_L$
preservation by the DNLS Hartree Hamiltonian, and close the
two-loop anomaly on the Hodge complex. The quartic coupling and
top-Yukawa correction are reported as HCSM-native candidates
matching observation to sub-percent accuracy but not derived
from substrate dynamics.
DC.identifier: HCSM-40-2026-09

B.2 BibTeX for Automated Indexing

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@article{PreschuttiHCSM40,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-40}: The Higgs Mechanism},  
  journal = {HCSM White Paper Series},  
  year = {2026},  
  month = {September},  
  note = {ORCID: 0009-0004-5445-1744},  
  keywords = {Hodge Complex Standard Model, HCSM, Higgs mechanism,  
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    DNLS Hartree,  $su(2)_L$  preservation, two-loop anomaly},  
  abstract = {We derive the Higgs doublet as the complexified B1  
    isotypic of  $\Omega^0$  at  $\lambda=4$ , derive the physical weak  
    direction from the vacuum zero state, prove exact  $su(2)_L$   
    preservation by the DNLS Hartree Hamiltonian, and close the  
    two-loop anomaly on the Hodge complex. The quartic coupling  
    and top-Yukawa correction are reported as HCSM-native  
    candidates matching observation to sub-percent accuracy but  
    not derived from substrate dynamics.}  
}
```